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# Lab: Stream Logs from EC2 to Amazon OpenSearch using Fluent Bit

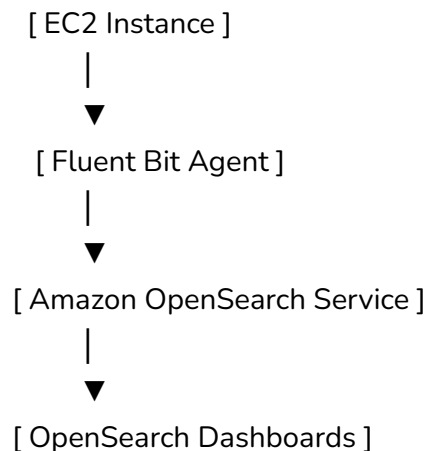
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## Objective

Stream application or system logs from an Amazon EC2 instance to Amazon OpenSearch Service and visualize them using OpenSearch Dashboards.

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## Architecture



## Pre-requisites

- ✓ Amazon OpenSearch Domain
- ✓ Amazon EC2 Instance (Amazon Linux 2 or Ubuntu)
- ✓ IAM Role with permissions to write to OpenSearch
- ✓ VPC endpoint or public endpoint for OpenSearch

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## Step-by-Step Instructions

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### Step 1: Create OpenSearch Domain

1. Go to **Amazon OpenSearch Service** → **Create Domain**
2. Choose:
  - **Deployment Type:** Development (for test)
  - **Engine:** OpenSearch 2.x
  - **Instance type:** `t3.small.search`
  - Enable **Fine-Grained Access Control**
  - Enable **Public access** or configure **VPC access**
3. Set access policy (IAM or fine-grained role)

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### Step 2: Launch EC2 Instance

1. Launch an **Amazon Linux 2** EC2 instance

Attach an IAM role that allows access to OpenSearch:

```
{  
  "Effect": "Allow",  
  "Action": [  
    "es:ESHttpPut",  
    "es:ESHttpPost"
```

```
],  
  "Resource": "arn:aws:es:region:account-id:domain/your-domain/*"  
}
```

2.

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### ✅ Step 3: Install Fluent Bit on EC2

```
sudo yum install -y amazon-linux-extras  
sudo amazon-linux-extras enable epel  
sudo yum install -y fluent-bit
```

Check version:

```
fluent-bit --version
```

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### ✅ Step 4: Configure Fluent Bit to Send Logs to OpenSearch

Edit the Fluent Bit config file:

```
sudo nano /etc/fluent-bit/fluent-bit.conf
```

Replace contents with:

```
[SERVICE]
```

```
  Flush      1  
  Log_Level  info
```

```
[INPUT]
```

```
  Name      tail  
  Path      /var/log/messages  
  Tag       ec2.logs  
  Refresh_Interval 5
```

#### [OUTPUT]

|            |                                   |
|------------|-----------------------------------|
| Name       | es                                |
| Match      | ec2.logs                          |
| Host       | <your-opensearch-domain-endpoint> |
| Port       | 443                               |
| Index      | ec2-logs                          |
| Type       | _doc                              |
| AWS_Auth   | On                                |
| AWS_Region | <your-region>                     |
| tls        | On                                |

Replace:

- <your-opensearch-domain-endpoint> with the domain (without https://)
- <your-region> (e.g., us-east-1)

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### ✓ Step 5: Start Fluent Bit

```
sudo systemctl start fluent-bit
sudo systemctl enable fluent-bit
```

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### ✓ Step 6: Validate Log Ingestion

- Go to **OpenSearch Dashboards**
- Navigate to **Discover**
- Create index pattern: `ec2-logs*`
- Explore logs streamed from `/var/log/messages`

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## Optional Enhancements

| Feature         | How                                                         |
|-----------------|-------------------------------------------------------------|
| Custom App Logs | Tail <code>/var/log/myapp.log</code> instead of system logs |
| Add Filters     | Use <code>[FILTER]</code> in Fluent Bit to enrich data      |
| Transform Logs  | Add JSON parsing for structured logs                        |
| Secure Traffic  | Use VPC endpoint + node-to-node encryption                  |

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## Sample Log Output (In OpenSearch)

```
{  
  "log": "Jul 29 17:12:54 ip-172-31-1-1 systemd: Starting Session 1...",  
  "source": "/var/log/messages",  
  "tag": "ec2.logs"  
}
```

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## Lab Completion Checklist

-  OpenSearch Domain
  -  Fluent Bit installed and configured
  -  Logs successfully streaming
  -  OpenSearch Dashboards visualizing logs
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